

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GPH 311 — Past Questions & Materials
by Renance

Course: GPH 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

This study pack was compiled and restructured by Renance for GPH 311 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 2 — GPH 311 Electrical Exploration Methods

Level 300 • First Semester • PastQuestion

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: DEPARTMENT OF EARTH SCIENCES

FIRST SEMESTER: B.Sc. (GEOPHYSICS) EXAMINATION 2024/2025 SESSION

COURSE: GPH 311(2 Units) ELECTRICAL EXPLORATION METHODS. TIME: 2 : First

INSTRUCTION: Answer question 1 and one question from each section

1a) Briefly, name and discuss one horizontal profiling (electrical resistivity tomography) technique for shallow

depth investigation \ J marks

1b) A client requires a detailed blue print to drill a borehole for ground water development in the past graduate

building

(i) Name and explain the appropriate field technique required for the geophysical survey

_ 8 marks

~~ {ii)- List two precautions that will enhance the acquisition process 2 marks

(iii) | Name two safety measures for the acquisition process 2 marks

__ 1c). State the law that governs the-resistivity-data acquisition process ~ 3 marks

SECTION A: Answer ONE Question from this Section

- 2a} If the potential at a point due to current flow is given by. — ~ a

iz

To

. = 3 derive the resistivity equation for a four electrode array system. /5 marks

Ur

2b) Briefly explain how the resistivity of rock is affected by its mineral content and porosity

is 6 marks

2c). What is the implication of increasing the current spread in field acquisition process using Schlumberger

technique f marks

3a) Differentiate between apparent resistivity and true resistivity 5 marks

3b) Explain briefly, why the resistivity of sedimentary facies in FUPRE health center is higher than the same

sedimentary facies in the coastal swamp of Ubeji town 3 marks

3c) Derive the resistivity equation for the Schlumberger configuration 15 marks

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SECTION B: Answer ONE question from this section '9

i (a) What are the origin of Spontaneous Potential (SP) in the ground? How would you minimize the electrochemical reaction between electrodes in spontaneous potential survey? 10 marks

(6) Briefly describe the Total Field Method of spontaneous potential survey. 1/5 marks

2(a) What do you understand by the term Chargeability in Induced Polarization? 5 marks

ye ety soelin the frequency domain of Induced Polarization. /0 marks

(bli) Describe the square-wave-in chargeable ground, Use a sketch for the description. Explain this relation

V

M=—. 10 marks c

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Paper 2 of 2 — GPH 311 ELECTRICAL EXPLORATION METHODS

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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DEPARTMENT OF EARTH SCIENCES

FIRST SEMESTER: B.Sc. (GEOPHYSICS) EXAMINATION 2024/2025 SESSION

COURSE: GPH 311(2 Units) ELECTRICAL EXPLORATION METHODS. TIME: 2 : Kirs

INSTRUCTION: Answer question 1 and one question from each section

1a) Briefly, name and discuss one horizontal profiling (electrical resistivity tomography) technique for shallow

depth investigation

5 marks

1b) A client requires a detailed blue print to drill a borehole for ground water development in the pest graduate

building

(1) Name and explain the appropriate field technique required for the geophysical survey

& marks

(ii)~ List two precautions that will enhance the acquisition process 2 marks

(iii) | Name two safety measures for the acquisition process 2 marks

1c) _ State the law that governs-the resistivity data acquisition process — 3 marks

SECTION A: Answer ONE Question from this Section

2a) If the potential at a point due to current flow is given by." Le

ip

a Sor derive the resistivity equation for a four elegi:ode array system. J/5 marks

2b) Briefly explain how the resistivity of rock is affected by its mineral content and porosity

6 marks

2c). What is the implication of increasing the current spr

vad in field acquisition process using Schiumberger

technique

1 marks

5a) Differentiate between apparent resistivity and true resistivity Smtarks

3b) Explain briefly, why the resistivity of sedimentary facie in FUPRE health center ts higher than the same

sedimentary facie in ihe coastal swamp of Ubeji town

5 marks

3c) Derive the reistivity equation for the Schlumberger configuration 15 marks

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SECTION B: Answer ONE question from this section 2

1 (a) What are the origin of Spontaneous Potential (SP) in the ground? How would you minimize the electrochemical reaction between electrodes in spontaneous potential survey? /0 marks

(6) Briefly describe the Total Field Method of spontaneous potential survey. 175 marks

2(a) What do you understand by the term Chargeability in Induced Polarization? 5 marks

(bi) Briefly explain the frequency domain of Induced Polarization. 10 marks

(bii) Describe the square-wave-in chargeable ground, Use

V_y

$M = \frac{V_y}{V_0}$. 10 marks ; bo si aaa

V_0 Cy vs ee em / >.

a sketch for the description. Explain this relation

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